

C++ + VS Code

(Quick Setup)

1. Install Visual Studio Code

Download and install:

[Visual Studio Code](#)

2. Install C++ Compiler

Download and install:

[MSYS2](#)

Open **MSYS2 UCRT64** and run:

```
pacman -Syu
```

Restart MSYS2 if requested, then run:

```
pacman -S --needed mingw-w64-ucrt-x86_64-gcc
```

3. Add Compiler to PATH

Add this folder to **Windows Environment Variables → Path**:

```
C:\msys64\ucrt64\bin
```

4. Check Installation

Open PowerShell and run:

```
g++ --version
```

If you see something like:

```
g++ (MinGW-W64 ...) 13.x.x
```

✅ C++ is ready.

5. Install C++ Extension

In VS Code: Extensions (Ctrl + Shift + X) → search C/C++ → Install

Publisher: **Microsoft**

6. Create Your First Program

Create a folder:

```
C:\CPP
```

Open it in VS Code and create a file named `main.cpp`:

```
main.cpp
```

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Write:

```
#include <iostream>

int main()
{
    std::cout << "Hello, C++!" << std::endl;

    return 0;
}
```

7. Compile and Run

Open: **Terminal** → **New Terminal**

Compile:

```
g++ -std=c++20 main.cpp -o main.exe
```

Run:

```
.\main.exe
```

You should see:

```
Hello, C++!
```

☒ **Your C++ environment is ready!**

VS Code + C++ Extension + MinGW-w64/G++ + C++20

IDE, Compiler ...

Basic → Structured → OOP → Data Structure → Design Pattern

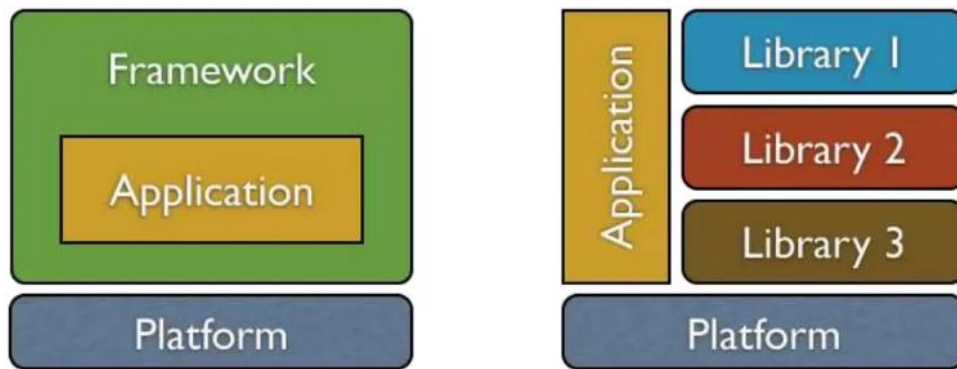
OOP

Class, Object, Access Modifiers, (private, public, protected), Constructors (constructors + getters/setters), Inheritance, Abstraction, Encapsulation, Polymorphism, Composition

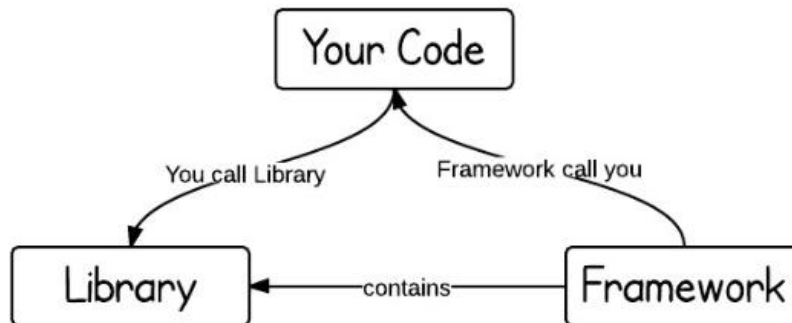
Module, Package, Library, Framework

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<https://www.slideshare.net/technical-overwiev>



Problems could be solved with:

Programming → Machine Learning → Deep Learning → ...?